

Целые точки и выпуклые тела

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The geometry of numbers is convex geometry in which volume is measured in units of the lattice. A single theorem of Minkowski yields Lagrange's four-square theorem, Dirichlet's approximations and the bound for the shortest vector of a lattice.

Definition 1. A lattice $\Lambda \subseteq \mathbb{R}^n$ is the set of integer combinations of a basis; $\det \Lambda$ is the volume of a fundamental parallelepiped.

Theorem 1 (Blichfeldt 1914). If $|S| > \det \Lambda$ for a measurable S , then S contains two distinct points whose difference lies in Λ .

Theorem 2 (Minkowski 1896). If K is convex, symmetric and $|K| > 2^n \det \Lambda$, then K contains a nonzero point of Λ ; for compact K the inequality may be non-strict.

Theorem 3 (Pick 1899). A lattice polygon with I interior and B boundary lattice points has area $I + B/2 - 1$.

1. Prove Blichfeldt's theorem and deduce Minkowski's theorem. (*Hint: apply Blichfeldt to $\frac{1}{2}K$.*)
2. Deduce from Minkowski's theorem that every prime $p \equiv 1 \pmod{4}$ is a sum of two squares, and that every positive integer is a sum of four squares.
3. Prove Pick's theorem and show that no analogue holds in $\mathbb{R}^3$: compute volumes and lattice point counts for the Reeve tetrahedra $T_k = \text{conv}\{0, e_1, e_2, e_1 + e_2 + ke_3\}$.
4. (classical) Prove that no equilateral triangle has all three vertices in $\mathbb{Z}^2$, and determine all regular polygons that can be inscribed in the

lattice $\mathbb{Z}^2$. (*Hint: the area of a lattice polygon is rational, while for an equilateral triangle it equals $\frac{\sqrt{3}}{4}a^2$.*)

5. (Mexmat, 2001) Find all convex equiangular polygons whose vertices are points of $\mathbb{Z}^2$.